

# **Year 12**

## **Mathematics**

### **Trial HSC Examination**

### **2014**

#### **General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- In Questions 11 - 16, show relevant mathematical reasoning and/or calculations

#### **Total marks – 100**

##### **Section I**

#### **10 marks**

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

##### **Section II**

#### **90 marks**

- Attempt Questions 11-16
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your name and "N/A" on the front cover
- Allow about 2 hours 45 minutes for this section

**DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM**

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## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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- 1 What is the primitive of  $\frac{2}{x} - \cos x$ ?
- (A)  $-\frac{2}{x^2} + \sin x + C$
- (B)  $-\frac{2}{x^2} - \sin x + C$
- (C)  $2 \ln x + \sin x + C$
- (D)  $2 \ln x - \sin x + C$
- 2 What are the values of  $x$  for which  $|4 - 3x| < 13$ ?
- (A)  $x < -3$  and  $x < \frac{17}{3}$
- (B)  $x > -3$  and  $x > \frac{17}{3}$
- (C)  $x > -3$  and  $x < \frac{17}{3}$
- (D)  $x < -3$  and  $x > \frac{17}{3}$
- 3 What is the simultaneous solution to the equations  $2x + y = 7$  and  $x - 2y = 1$ ?
- (A)  $x = 3$  and  $y = 1$
- (B)  $x = -1$  and  $y = 9$
- (C)  $x = 2$  and  $y = 3$
- (D)  $x = 5$  and  $y = 1$

- 4 Factorise  $2x^2 - 7x - 15$
- (A)  $(2x-3)(x-5)$
- (B)  $(2x+3)(x-5)$
- (C)  $(2x-5)(x-3)$
- (D)  $(2x+5)(x-3)$
- 5 The value of  $\frac{5.79+0.55}{\sqrt{4.32-3.28}}$  is closest to:
- (A) 4
- (B) 6
- (C) 9
- (D) 10
- 6 What are the values of  $p$  and  $q$  given  $(3\sqrt{12} + \sqrt{75})(2 + \sqrt{48}) = p + q\sqrt{3}$ ?
- (A)  $p = 132$  and  $q = 15$
- (B)  $p = 396$  and  $q = 15$
- (C)  $p = 132$  and  $q = 22$
- (D)  $p = 396$  and  $q = 22$
- 7 The line  $6x - ky = 8$  passes through the point  $(3, 2)$ . What is the value of  $k$ ?
- (A) -13
- (B) -5
- (C) 5
- (D) 15

- 8 The semi-circle  $y = \sqrt{4 - x^2}$  is rotated about the  $x$ -axis. Which of the following expressions is correct for the volume of the solid of revolution?
- (A)  $V = \pi \int_0^2 (4 - x^2) dx$
- (B)  $V = 2\pi \int_0^2 (4 - x^2) dx$
- (C)  $V = \pi \int_0^2 (4 - y^2) dy$
- (D)  $V = 2\pi \int_0^2 (4 - y^2) dy$
- 9 A circle has the equation  $4x^2 - 4x + 4y^2 + 24y + 21 = 0$ . What is the radius and centre?
- (A) Centre  $\left(\frac{1}{2}, -3\right)$  and radius of 2.
- (B) Centre  $\left(\frac{1}{2}, 3\right)$  and radius of 2.
- (C) Centre  $\left(\frac{1}{2}, -3\right)$  and radius of 4.
- (D) Centre  $\left(\frac{1}{2}, 3\right)$  and radius of 4.
- 10 An infinite geometric series has a first term of 12 and a limiting sum of 15. What is the common ratio?
- (A)  $\frac{1}{5}$
- (B)  $\frac{1}{4}$
- (C)  $\frac{1}{3}$
- (D)  $\frac{1}{2}$

## Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

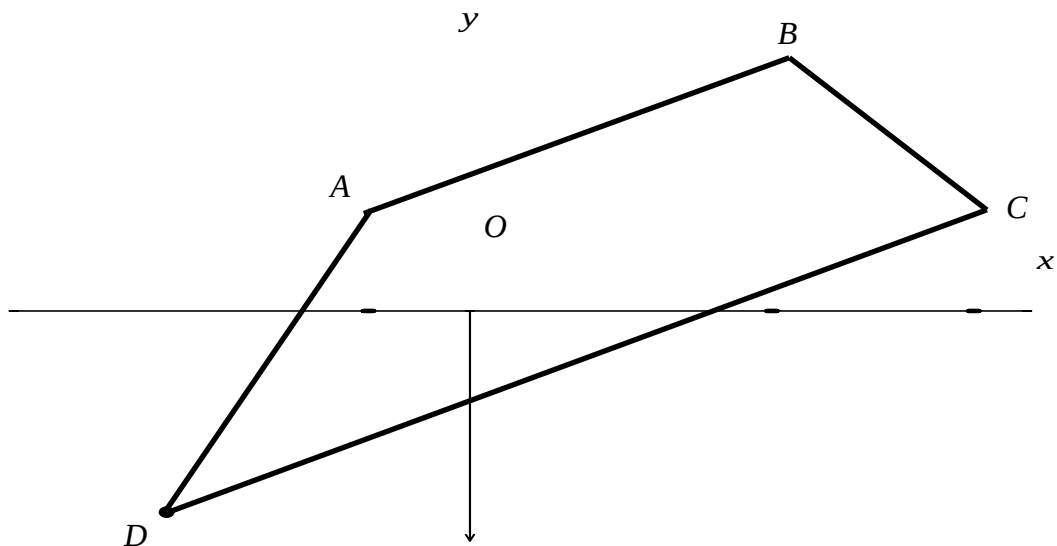
Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

### Question 11 (15 marks)

Marks

- (a) The diagram shows the points  $A(-1,1)$ ,  $B(3,6)$ ,  $C(5,1)$  and  $D(-3,-9)$ .



- |       |                                                             |   |
|-------|-------------------------------------------------------------|---|
| (i)   | Find the coordinates of $E$ , the midpoint of $DC$ .        | 1 |
| (ii)  | Show that the equation of $BE$ is $5x - y - 9 = 0$ .        | 2 |
| (iii) | Find the perpendicular distance from $A$ to the line $BE$ . | 2 |
| (iv)  | Show that $ABED$ is a parallelogram.                        | 2 |
| (v)   | Find the area of $ABED$ .                                   | 1 |

Question 11 continues over the page

- (b) Find the equation of the tangent to the curve  $y = \log_e x - 1$  at the point  $(e, 0)$ . 2
- (c) The equation of a parabola is given by  $y = x^2 - 2x + 5$ .
- (i) Find the coordinates of its vertex. 2
- (ii) State the focal length of the parabola. 1
- (iii) Find the equation of the normal at the point  $P(2, 5)$ . 2

**End of Question 11**

**Question 12 (15 marks)** Use a SEPARATE writing booklet**Marks**

- (a) There are 200 tickets sold in a raffle with only two prizes. These tickets are placed in a bag and two are drawn, one at a time. Once a ticket is drawn it is not placed back in the bag. One boy bought 3 tickets. calculate the probability the boy wins:

- |       |                        |   |
|-------|------------------------|---|
| (i)   | First prize.           | 1 |
| (ii)  | Both prizes.           | 1 |
| (iii) | The second prize only. | 1 |
| (iv)  | No prize at all.       | 1 |

- (b) Differentiate with respect to  $x$ .

- |      |                      |   |
|------|----------------------|---|
| (i)  | $e^{3x} \tan x$      | 2 |
| (ii) | $\frac{\sin x}{5-x}$ | 2 |

- (c) Evaluate

- |      |                                      |   |
|------|--------------------------------------|---|
| (i)  | $\int \frac{1}{1-2x} dx$             | 2 |
| (ii) | $\int_0^{\pi} \sec^2 \frac{x}{3} dx$ | 2 |

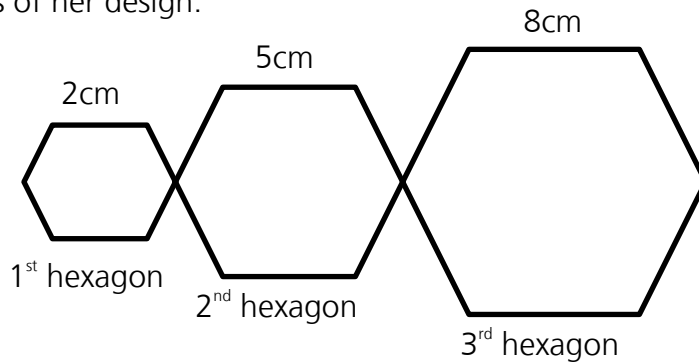
- (d) The roots of the equation  $2x^2 - x - 15 = 0$  are  $\alpha$  and  $\beta$ . Find the value of:

- |       |                                      |   |
|-------|--------------------------------------|---|
| (i)   | $\alpha + \beta$                     | 1 |
| (ii)  | $\alpha\beta$                        | 1 |
| (iii) | $\frac{1}{\alpha} + \frac{1}{\beta}$ | 1 |

**End of Question 12**

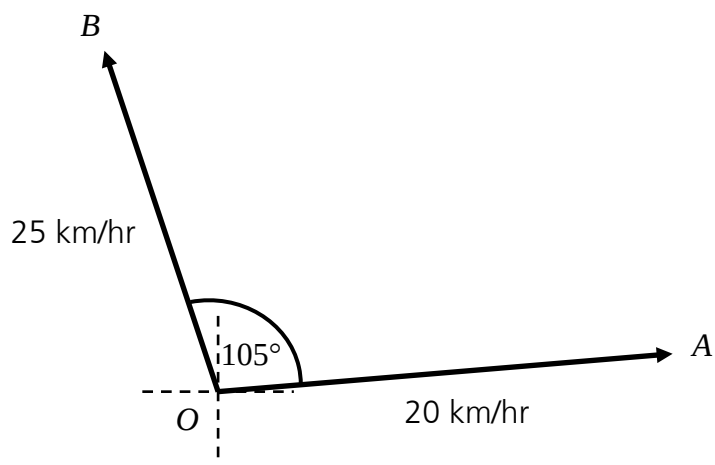
**Question 13 (15 marks)** Use a SEPARATE writing booklet**Marks**

- (a) Melanie is using wire to construct a geometrical design which consists of  $n$  regular hexagons with sides 2cm, 5cm, 8cm and so on going up by the same amount each hexagon. The diagram below shows the first 3 hexagons of her design.



- |       |                                                                                                        |   |
|-------|--------------------------------------------------------------------------------------------------------|---|
| (i)   | Find the perimeter of the $n$ th hexagon.                                                              | 1 |
| (ii)  | Show that the total length of the wire is $L = 9n^2 + 3n$ .                                            | 2 |
| (iii) | If the total length of the wire is 6 metres, find the number of hexagons that Melanie has constructed. | 2 |
- (b) Let  $f(x) = x^3 - 3x^2 - 9x + 22$
- |       |                                                                                                                                   |   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|---|
| (i)   | Show that $f''(x) = 6x - 6$                                                                                                       | 1 |
| (ii)  | Find the coordinates of the stationary points on $y = f(x)$ and determine their nature.                                           | 2 |
| (iii) | Find the coordinates of the point(s) of inflexion.                                                                                | 2 |
| (iv)  | Sketch the graph of $y = f(x)$ , indicating where the curve meets the $y$ -axis, stationary points and the point(s) of inflexion. | 2 |
| (v)   | For what values of $x$ is the graph of $y = f(x)$ concave down?                                                                   | 1 |

- (c) Alex and Bella leave from point  $O$  at the same time. Alex travels at  $20\text{km/h}$  along a straight road in the direction  $085^\circ$ . Bella travels at  $25\text{km/h}$  along another straight road in the direction  $340^\circ$ .

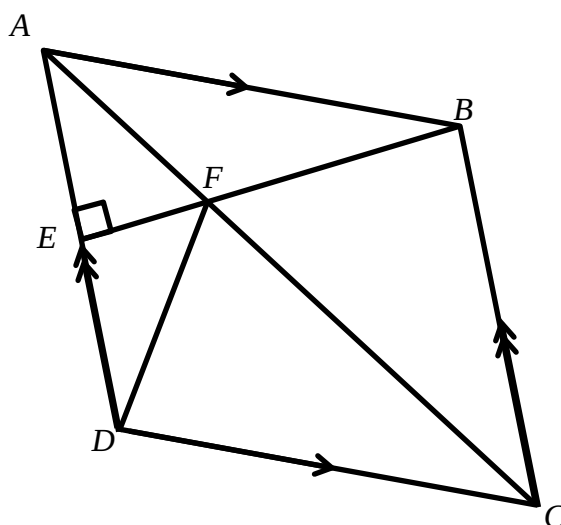


Find the distance Alex and Bella are apart to the nearest kilometre after two hours.

2

**End of Question 13**

(a)

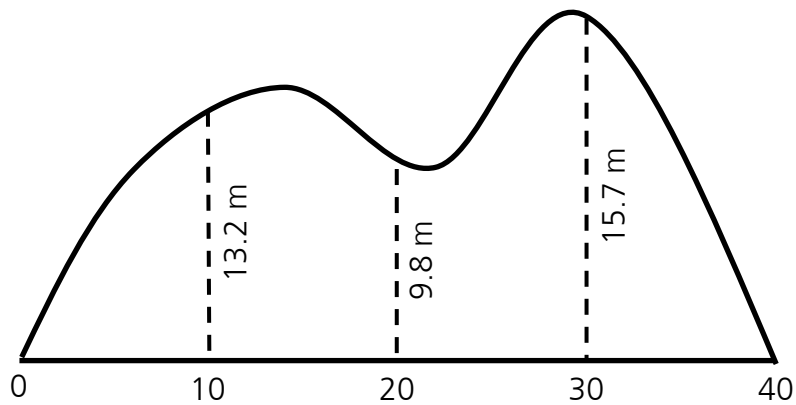


$ABCD$  is a rhombus,  $BE$  is perpendicular to  $AD$  and intersects  $AC$  at  $F$ .  
Copy or trace the diagram into your writing booklet.

- |       |                                                         |   |
|-------|---------------------------------------------------------|---|
| (i)   | Explain why $\angle BCA = \angle DCA$ .                 | 1 |
| (ii)  | Prove that the triangles $BFC$ and $DFC$ are congruent. | 3 |
| (iii) | Show that $\angle FBC$ is a right angle.                | 1 |
| (iv)  | Hence, or otherwise, find the size of $\angle FDC$ .    | 1 |
- (b) A scientist grows the number of bacteria according to the equation
- $$N(t) = Ae^{0.15t}$$
- Where  $t$  is measured in days and  $A$  is a constant.
- |       |                                                                                                                                  |   |
|-------|----------------------------------------------------------------------------------------------------------------------------------|---|
| (i)   | Show that the number of bacteria increases at a rate proportional to the number present.                                         | 2 |
| (ii)  | When $t = 3$ the number of bacteria was estimated at $1.5 \times 10^8$ . Evaluate $A$ . Answer correct to 2 significant figures. | 1 |
| (iii) | The number of bacteria doubles every $x$ days. Find $x$ . Answer correct to 1 decimal place.                                     | 2 |

**Question 14 continues over the page**

- (c) During a survey the area of an irregular headland was to be found. Measurements of the area were noted on the diagram below.



Use Simpson's rule with 5 function values to estimate the area of the headland. Answer correct to the nearest square metre.

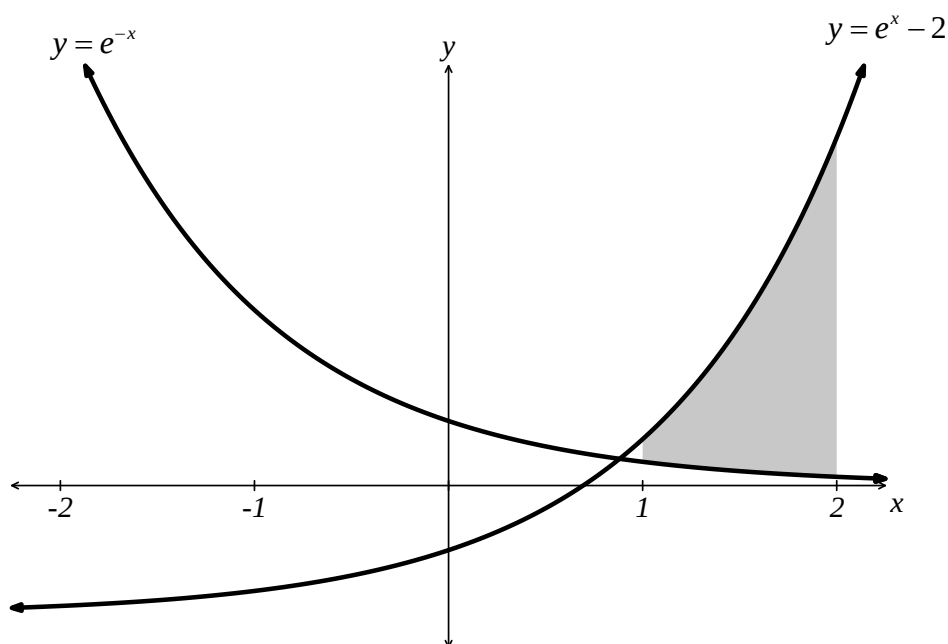
2

- (d) Solve the equation  $(\cos x + 2)(2 \cos x + 1) = 0$  in the domain  $0 \leq x \leq 2\pi$ .

2

**End of Question 14**

(a)



The diagram shows the graphs of  $y = e^x - 2$  and  $y = e^{-x}$ .

- (i) Find the area between the curves from  $x = 1$  to  $x = 2$ . Leave your answer in terms of  $e$ . 3
- (ii) Show that the curves intersect when  $e^{2x} - 2e^x - 1 = 0$ . 2
- (iii) Hence, using the substitution  $u = e^x$ , or otherwise, find the point of intersection of the curves. 2

- (b) Velocity of an object moving along the  $x$ -axis is given by  
 $v = 2\sin t + 1$  for  $0 \leq t \leq 2\pi$

Where  $v$  is measured in metres per second and  $t$  in seconds.

- |       |                                                                                        |   |
|-------|----------------------------------------------------------------------------------------|---|
| (i)   | When is the object at rest?                                                            | 2 |
| (ii)  | Sketch the graph of $v$ as a function of $t$ for $0 \leq t \leq 2\pi$                  | 2 |
| (iii) | Find the maximum velocity of the object for this period.                               | 1 |
| (iv)  | When is the object travelling in the negative direction during this period?            | 1 |
| (v)   | Calculate the total distance travelled by the object in the period $0 \leq t \leq \pi$ | 2 |

**End of Question 15**

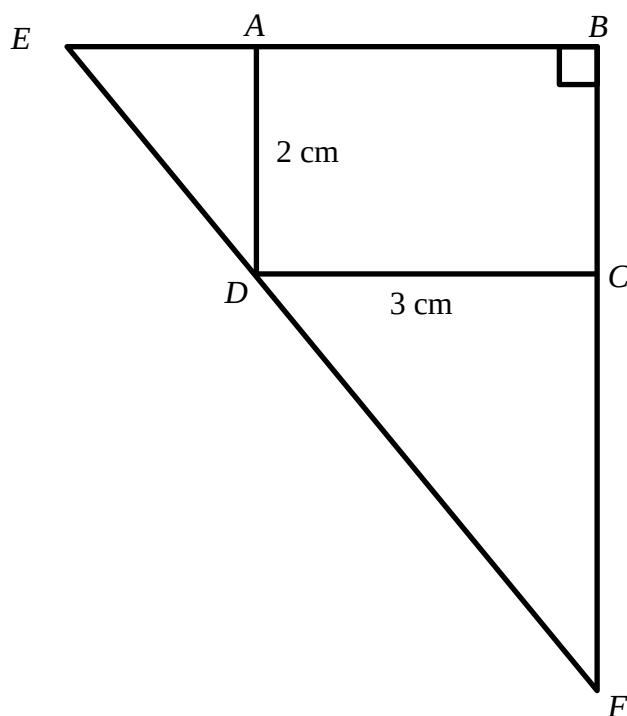
**Question 16 (15 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) George is saving for a holiday. He opens a savings account with an interest rate of 0.4% per month compounded monthly at the end of each month. George decides to deposit \$450 into the account on the first day of each month. He makes his first deposit on the 1<sup>st</sup> of January 2012 and his last on the 1<sup>st</sup> of December 2014. George withdraws the entire amount, plus interest, immediately after his final interest payment on the 31<sup>st</sup> December 2014.
- (i) How much in total did George deposit into his savings account over this period? 1
- (ii) How much did George withdraw from his account on the 31<sup>st</sup> December 2014? Answer correct to the nearest dollar. 3
- (iii) George's holiday is postponed due to family illness. He decides to deposit \$12 000 into a different account with an interest rate of 5% p.a. compounded quarterly for 2 years. How much will George receive at the end of the investment period from this \$12000 investment? Answer correct to the nearest dollar. 2

**Question 16 continues over the page**

(b)



$ABCD$  is a rectangle with  $CD = 3$  cm and  $AD = 2$  cm.  $F$  and  $E$  lie on the lines  $BC$  and  $BA$ , so that,  $F$ ,  $D$  and  $E$  are collinear. Let  $CF = x$  cm and  $AE = y$  cm.

- |       |                                                                                                  |   |
|-------|--------------------------------------------------------------------------------------------------|---|
| (i)   | Show that $\triangle FCD$ and $\triangle DAE$ are similar.                                       | 3 |
| (ii)  | Show that $xy = 6$                                                                               | 1 |
| (iii) | Show that the area ( $A$ ) of $\triangle FBE$ is given by $A = 6 + \frac{3}{2}x + \frac{6}{x}$ . | 2 |
| (iv)  | Find the height and base of $\triangle FBE$ with minimum area. Justify your answer.              | 3 |

**End of Examination**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

Note  $\ln x = \log_e x, \quad x > 0$

## Mathematics

## Section I Multiple-Choice Answer Sheet

1      A ☐      B ☐      C ☐      D ☐2      A ☐      B ☐      C ☐      D ☐3      A ☐      B ☐      C ☐      D ☐4      A ☐      B ☐      C ☐      D ☐5      A ☐      B ☐      C ☐      D ☐6      A ☐      B ☐      C ☐      D ☐7      A ☐      B ☐      C ☐      D ☐8      A ☐      B ☐      C ☐      D ☐9      A ☐      B ☐      C ☐      D ☐10     A ☐      B ☐      C ☐      D ☐

## Mathematics

## Section I Multiple-Choice Answer Sheet

- |    |                                    |                                    |                                    |                                    |
|----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1  | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 2  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 3  | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 4  | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 5  | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 6  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 7  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 8  | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 9  | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 10 | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |

Year 12 2U Trial HSC 2014

①

1)  $\int \frac{2}{x} - \cos x \, dx = 2 \ln x - \sin x + c.$

**D**

2)  $|4 - 3x| < 13$

$$4 - 3x < 13$$

$$-9 < 3x$$

$$x > -3$$

$$4 - 3x > -13$$

$$17 > 3x$$

$$x < \frac{17}{3}$$

**C**

3)  $2x + y = 7$  ... ①  
 $x - 2y = 1$  ... ②

①  $4x + 2y = 14$   
 $x - 2y = 1$

$$5x = 15$$

$$x = 3$$

$$y = 1$$

**A**

4)  $2x^2 - 7x - 15 = (2x + 3)(x - 5)$  **B**

5) **B**

6)  $(3\sqrt{12} + \sqrt{75})(2 + \sqrt{48}) = (6\sqrt{3} + 5\sqrt{3})(2 + 4\sqrt{3})$   
 $= 11\sqrt{3}(2 + 4\sqrt{3})$   
 $= 22\sqrt{3} + 132,$

$p = 132$   
 $q = 22,$

**C**

2

$$7) 6(3) - 2k = 8.$$

$$2k = 10$$
$$k = 5$$

C

$$8) V = 2\pi \int_0^2 4 - x^2 dx. \quad \underline{B}$$

$$9) 4x^2 - 4x + 4y^2 + 24y = -21$$

$$x^2 - x + y^2 + 6y = -\frac{21}{4}$$

$$x^2 - x + \frac{1}{4} + y^2 + 6y + 9 = -\frac{21}{4} + \frac{1}{4} + 9$$

$$(x - \frac{1}{2})^2 + (y + 3)^2 = 4$$

$$(\frac{1}{2}, -3) \quad R = 2$$

A

$$10) a = 12$$
$$S_n = 15$$

$$15 = \frac{12}{1-r}$$

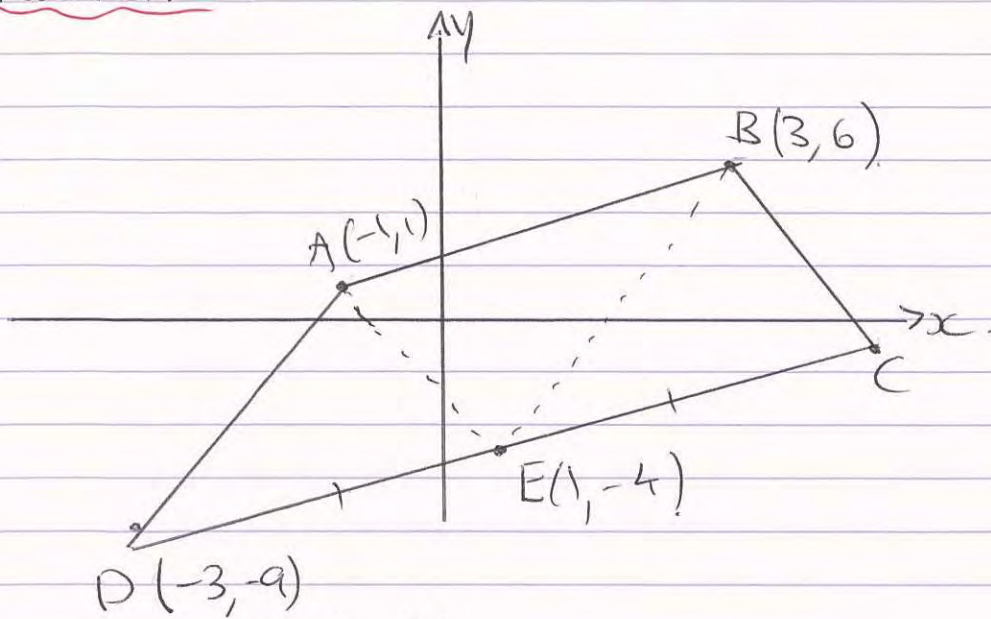
$$1-r = \frac{12}{15}$$

$$r = 1 - \frac{4}{5}$$

$$= \frac{1}{5}$$

A

(3)

Question 11

$$i) E\left(\frac{5-3}{2}, \frac{1-9}{2}\right)$$

$$E(1, -4)$$

$$ii) m_{BE} = \frac{6+4}{3-1} \\ = 5$$

$$y+4 = 5(x-1)$$

$$y+4 = 5x-5$$

$$5x - y - 9 = 0$$

$$iii) M_{AE}\left(\frac{-1+1}{2}, \frac{1-4}{2}\right)$$

$$M_{BD}\left(\frac{3-3}{2}, \frac{6-9}{2}\right)$$

$$M_{AE}\left(0, -\frac{3}{2}\right)$$

$$M_{BD}\left(0, -\frac{3}{2}\right)$$

∴ ABED is a parallelogram (Diagonals bisect each other)

(iii)

$$\text{iv) } A(-1, 1)$$

$$d = \frac{|5(-1) - (1) - 9|}{\sqrt{5^2 + 1^2}}$$

$$= \frac{15}{\sqrt{26}}$$

$$\text{v) } BE = \sqrt{(3-1)^2 + (6+4)^2}$$

$$= \sqrt{104}$$

$$= 2\sqrt{26}$$

$$A = 2\sqrt{26} \times \frac{15}{\sqrt{26}}$$

$$= \underline{\underline{30 \text{ u}^2}}$$

$$\text{b) } y = \ln x - 1$$

$$y' = \frac{1}{x}$$

$$\text{@ } x = e, y' = ?$$

$$y' = \frac{1}{e}$$

$$\text{At } x = e$$

$$y - 0 = \frac{1}{e}(x - e)$$

$$y = \frac{x}{e} - 1 \quad \text{OR} \quad x - ey - e = 0$$

5

$$c i) \quad y - 5 = x^2 - 2x$$

$$y - 5 + 1 = x^2 - 2x + 1$$

$$y - 4 = (x - 1)^2$$

$$V = (1, 4)$$

$$ii) \quad a = \frac{1}{4}$$

$$iii) \quad y' = 2x - 2$$

$$@ x = 2,$$

$$y' = 2$$

$$m_N = -\frac{1}{2}$$

$$y - 5 = -\frac{1}{2}(x - 2)$$

$$2y - 10 = -x + 2$$

$$x + \underline{2y - 12} = 0$$

$$\text{or } y = 6 - \frac{x}{2}$$

(6)

### Question 12

$$\text{ai) } P(1\text{st}) = \frac{3}{200} \quad \checkmark$$

$$\begin{aligned} \text{ii) } P(1\text{st} + 2\text{nd}) &= \frac{3}{200} \times \frac{2}{199} \quad \checkmark \\ &= \frac{3}{19900} \end{aligned}$$

$$\begin{aligned} \text{iii) } P(2\text{nd}) &= \frac{197}{200} \times \frac{3}{199} \quad \checkmark \\ &= \frac{591}{39800} \end{aligned}$$

$$\begin{aligned} \text{iv) } P(\text{none}) &= \frac{197}{200} \times \frac{196}{199} \quad \checkmark \\ &= \frac{9653}{9950} \end{aligned}$$

$$\begin{aligned} \text{bi) } \frac{d}{dx} (e^{3x} \tan x) &= 3e^{3x} \tan x + e^{3x} \cdot \sec^2 x. \quad \checkmark \checkmark \\ &= e^{3x} (3 \tan x + \sec^2 x) \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{d}{dx} \left( \frac{\sin x}{5-x} \right) &= \frac{\cos x (5-x) - \sin x \cdot (-1)}{(5-x)^2} \\ &= \frac{5 \cos x - x \cos x + \sin x}{(5-x)^2} \quad \checkmark \checkmark \end{aligned}$$

7

$$c i) \int \frac{1}{1-2x} dx = -\frac{1}{2} \ln(1-2x) + C \quad \checkmark \checkmark$$

$$\begin{aligned} ii) \int_0^{\pi} \sec^2 \frac{x}{3} dx &= \left[ 3 \tan \frac{x}{3} \right]_0^{\pi} \quad \checkmark \\ &= 3 \tan \frac{\pi}{3} \\ &= \underline{\underline{3\sqrt{3}}} \quad \checkmark \end{aligned}$$

$$d i) \alpha + \beta = \frac{1}{2} \quad \checkmark$$

$$ii) \alpha \beta = -\frac{15}{2} \quad \checkmark$$

$$\begin{aligned} iii) \frac{1}{\alpha} + \frac{1}{\beta} &= \frac{\alpha + \beta}{\alpha \beta} \\ &= \frac{\frac{1}{2}}{-\frac{15}{2}} \\ &= \underline{\underline{-\frac{1}{15}}} \end{aligned}$$

8

### Question 13

$$\begin{aligned} \text{ai)} \quad P &= 6(2 + (n-1) \times 3) \\ &= 6(2 + 3n - 3) \\ &= 18n - 6 \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad S_n &= \frac{n}{2} (18n - 6 + 2 \times 6) \\ &= \frac{n}{2} (18n + 6) \\ &= \underline{\underline{9n^2 + 3n}} \end{aligned}$$

$$\text{iii)} \quad 600 = 9n^2 + 3n$$

$$\begin{aligned} 3n^2 + n - 200 &= 0 \\ (3n + 25)(n - 8) &= 0 \end{aligned}$$

$$n = -\frac{25}{3}, 8$$

$$\underline{\underline{n = 8}}$$

$$\text{bi)} \quad f'(x) = 3x^2 - 6x - 9$$

$$f''(x) = 6x - 6$$

$$\text{i)} \quad \text{Stat pts @ } f'(x) = 0$$

$$0 = 3(x^2 - 2x - 3)$$

$$(x - 3)(x + 1) = 0$$

$$x = 3, -1$$

9

$$@x = -1, y = 27$$

$$y'' = -6 - 6 \\ = -12$$

$\therefore (-1, 27)$  is a max t.p. ✓

$$@x = 3, y = -5$$

$$y'' = 18 - 6 \\ = 12$$

$\therefore (3, -5)$  is a min t.p.  $(3, -12)$  ✓

ii) POI @  $y'' = 0$

$$0 = 6x - 6$$

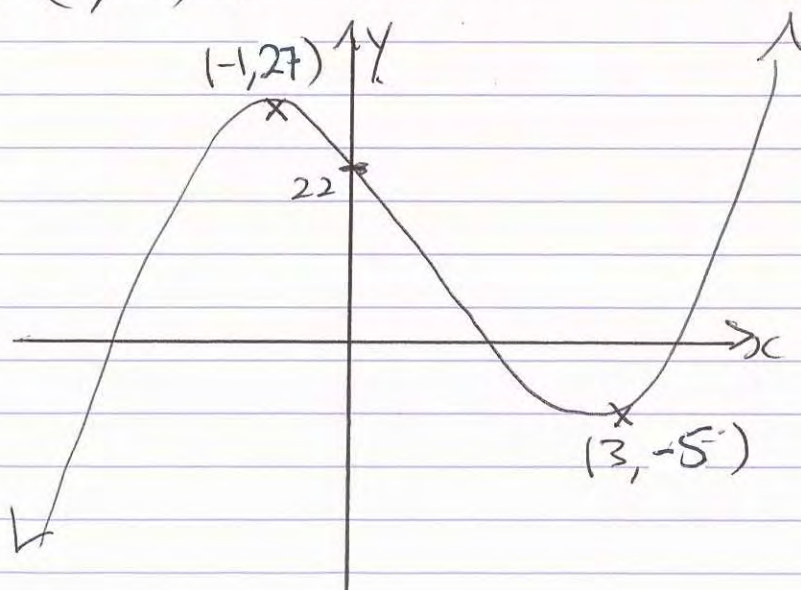
$$x = 1$$

Check

|     |   |   |   |
|-----|---|---|---|
| x   | 0 | 1 | 2 |
| y'' | - | 0 | + |

$\therefore (1, 11)$  is a POI ✓

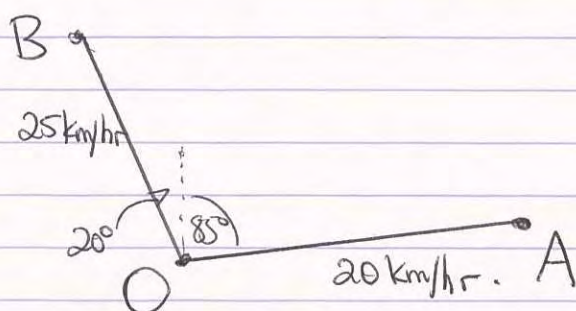
iv)



✓✓

v)  $x < 1$  ✓

c) ✓



$$\angle AOB = 20 + 85 \\ = 105^\circ$$

~~2)~~  $AO = 40 \text{ km}$   
 $BO = 50 \text{ km}$

$$AB^2 = 40^2 + 50^2 - 2 \times 40 \times 50 \times \cos 105^\circ \quad \checkmark \\ = 5135.28 \dots$$

$$AB = \underline{\underline{72 \text{ km}}} \quad \checkmark$$

(11)

Question 14

ai) Diagonals in a rhombus bisect the angles they pass through. ✓

ii) FC is common.

$$\angle BCA = \angle DCA \quad (\text{given (i)}) \quad \checkmark$$

$$BC = DC \quad (\text{equal sides in a rhombus}) \quad \checkmark$$

$$\therefore \triangle BFC \equiv \triangle DFC \quad (\text{SAS}) \quad \checkmark$$

iii)  $\angle FBC = 90^\circ$  (alt  $\angle$ 's on  $\parallel$  lines) ✓

iv)  $\angle FDC = 90^\circ$  (corres  $\angle$ 's in congruent  $\Delta$ 's) ✓

bi)  $N(t) = Ae^{0.15t}$

$$\frac{dN}{dt} = 0.15 \cdot Ae^{0.15t} \quad \checkmark$$

$$= 0.15N.$$

$$\frac{dN}{dt} \propto N \quad \checkmark$$

ii) @  $t=3$ ,  $N = 1.5 \times 10^8$

$$1.5 \times 10^8 = Ae^{0.45}$$

$$A = \frac{1.5 \times 10^8}{e^{0.45}}$$

$$= \underline{\underline{9.6 \times 10^7}} \text{ bacteria} \quad \checkmark$$

OR 960000000

(12)

iii)  $N=2$ .

$$2 = e^{0.15t}$$

$$\ln 2 = 0.15t$$

$$t = \frac{\ln 2}{0.15}$$

$$= 4.62$$

$$= 4.6 \text{ days}$$

c)

|    |   |      |      |      |    |
|----|---|------|------|------|----|
| x  | 0 | 10   | 20   | 30   | 40 |
| y  | 0 | 13.2 | 9.8  | 15.7 | 0  |
| w  | 1 | 4    | 2    | 4    | 1  |
| yw | 0 | 52.8 | 19.6 | 62.8 | 0  |

$$A \div \frac{10}{3} (52.8 + 19.6 + 62.8)$$

$$\div 450.7$$

$$\div \underline{\underline{451 \text{ m}^2}}$$

d)  $(\cos x + 2)(2\cos x + 1) = 0$

$$\cos x = -2, -\frac{1}{2}$$

$$\cos x \neq -2$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

Question 15

$$\text{ai.) } A = \int_1^2 e^x - 2 - e^{-x} dx. \quad \checkmark$$

$$= \left[ e^x - 2x + e^{-x} \right]_1^2 \quad \checkmark$$

$$= (e^2 - 4 + e^{-2}) - (e - 2 + e^{-1})$$

$$= \underline{\underline{e^2 - e - 2 - e^{-1} + e^{-2}}}. \quad \checkmark$$

$$\text{ii.) } \begin{aligned} y &= e^x - 2 \\ y &= e^{-x} \end{aligned}$$

$$e^x - 2 = e^{-x}$$

$$e^x - 2 = \frac{1}{e^x} \quad \checkmark$$

$$e^{2x} - 2e^x = 1$$

$$\underline{\underline{e^{2x} - 2e^x - 1 = 0}} \quad \checkmark$$

$$\text{iii.) } \dots$$

$$u^2 - 2u - 1 = 0$$

$$\{u = \frac{2 \pm \sqrt{4 + 4}}{2}$$

$$= 1 \pm \sqrt{2}$$

$$e^x = 1 + \sqrt{2}. \quad \checkmark$$

$$x = \ln(1 + \sqrt{2}) \stackrel{\text{or.}}{=} 0.881 \quad \checkmark \quad \therefore y = 0.414$$

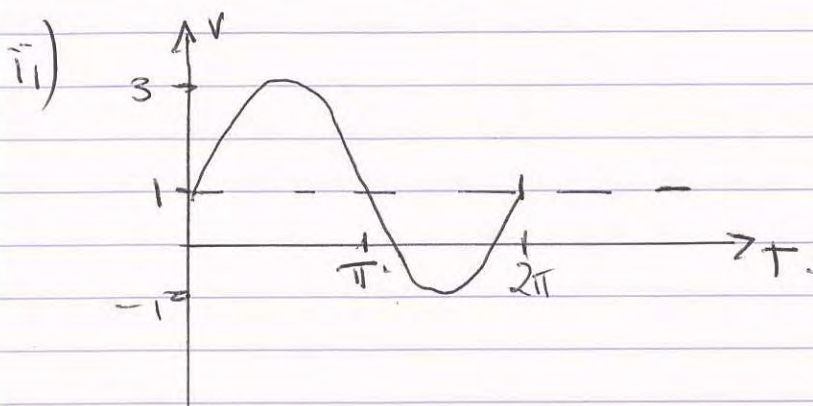
(14)

bi) @  $v=0$ ,  $t=?$

$$0 = 2\sin t + 1$$

$$\sin t = -\frac{1}{2}$$

$$t = \frac{7\pi}{6}, \frac{11\pi}{6} \text{ s. } \checkmark \checkmark$$



iii)  $v=3$   $\checkmark$

iv)  $\frac{7\pi}{6} < t < \frac{11\pi}{6}$

v)  $d = \int_0^{\pi} 2\sin t + 1 \, dt$   $\checkmark$

$$= [t - 2\cos t]_0^{\pi}$$

$$= (\pi - 2\cos \pi) - (0 - 2\cos 0)$$

$$= \pi + 2 + 2$$

$$= \underline{\underline{\pi + 4}} \quad \checkmark$$

## Question 16

ai)  $r = 0.004$

$$A = 450 \times 36$$

$$= \$15800 \rightarrow \$16200 \quad \checkmark$$

ii)  $A_1 = 450 \times 1.004^{36}$

$$A_2 = 450 \times 1.004^{35}$$

$$A_{36} = 450 \times 1.004$$

$$\text{Total} = 450 (1.004 + 1.004^2 + \dots + 1.004^{36})$$

$$= 450 \times 1.004 \frac{(1.004^{36} - 1)}{0.004} \quad \checkmark$$

$$= \$17456.70$$

$$= \$17457 \quad \checkmark$$

iii)  $r = 0.05 \div 4$   
 $= 0.0125$

$$A = 12000 (1.0125)^8 \quad \checkmark$$

$$= 13253.83$$

$$= \$13254 \quad \checkmark$$

(16)

b.i).  $\angle FCD = \angle DAE = 90^\circ$  ( $\angle$ 's on a straight line)

Let  $\angle FDC = x$

$$\angle ADE = 180 - 90 - x$$

$$= 90 - x \quad (\angle \text{'s})$$

$$\angle AED = 180 - 90 - (90 - x)$$

( $\angle$  sum of  $\Delta$ )

$$\angle FDC = \angle AED$$

$\therefore \Delta FCD \parallel \Delta DAE$

(equilateral)

$$\frac{CF}{AD} = \frac{CD}{AE}$$

$$\frac{x}{2} = \frac{3}{8}y$$

$$xy = 6$$

$$y = \frac{6}{x}$$

$$iii) A = \frac{1}{2} (BF) \times (BE)$$

$$= \frac{1}{2} (x+2)(y+3)$$

$$= \frac{1}{2} (xy + 3x + 2y + 6)$$

$$= \frac{1}{2} \left( x \cdot \frac{6}{x} + 3x + 2 \cdot \frac{6}{x} + 6 \right)$$

$$= \frac{1}{2} \left( 6 + 3x + \frac{12}{x} + 6 \right)$$

$$= \frac{1}{2} \left( 12 + 3x + \frac{12}{x} \right)$$

$$A = 6 + \frac{3}{2}x + \frac{6}{x}$$

$$\text{iv) } \frac{dA}{dx} = \frac{3}{2} - \frac{6}{x^2}$$

$$\text{Stat. pts @ } \frac{dA}{dx} = 0$$

$$0 = \frac{3}{2} - \frac{6}{x^2}$$

$$\frac{6}{x^2} = \frac{3}{2}$$

$$12 = 3x^2$$

$$x^2 = 4$$

$$x = 2 \quad \checkmark \quad x > 0$$

$$y = 3$$

Height is 4 cm  
Base is 6 cm  $\checkmark$

$$\frac{d^2A}{dx^2} = \frac{12}{x^3}$$

$$x = 2 \quad \frac{d^2A}{dx^2} > 0 \therefore \text{min. } \checkmark$$